

Dexter Carpenter

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Mechatronics Engineer

Overview

Mechatronics Engineer with a strong foundation in mechanical engineering and over a year of professional experience in industrial automation. I've developed a multidisciplinary skillset spanning mechanical design, electrical systems, and controls engineering, with a focus on rapidly delivering functional, scalable solutions in fast-moving environments. Through both industry experience and academic projects, I've taken on complex technical challenges and contributed to full system development from concept through implementation. I'm driven to continue growing as an engineer while delivering reliable and scalable solutions in the automation space.

Experience

Mechatronics Engineer, Timberline Automation & Control Co. (TAC) Dec. 2024 – Apr. 2026

- Designed and built integrated mechanical, electrical, and controls systems
- Developed CAD assemblies, control panels, and PLC software
- Supported projects from concept through commissioning

Control Panels & PLC Software for Lab0's 'Harvester'

Dec. 2024 – Apr. 2026

- Designed and owned electrical control panels for a robotic truck unloading system
- Authored complete electrical drawing packages
- Supported integration of dual 6-axis robotic arms with conveyor systems
- Architected PLC software using object-oriented frameworks
- Contributed to successful commissioning in a live distribution facility

CAD Vault Owner/Manager for Recycletek's Product Line

Apr. 2025 – Apr. 2026

- Managed a complex CAD vault spanning multiple interdependent products
- Maintained version control and ensured consistency across designs
- Served as the primary point of contact for design changes
- Produced manufacturing and assembly drawings

Project Lead, OPEnS Lab FloDar

Jan. 2021 – Nov. 2022

- Worked to continue the development of *FloDar*, an environmental sensor to monitor groundwater discharge and its properties.
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Projects

Rocket Air Brake Control System for OSU AIAA ESRA

Sept. 2023 – Jun. 2024

Co-developed a rocket air brake system including mechanical design, control algorithms, and simulation tools as part of a senior capstone project. Contributed to a fully integrated subsystem for a competitive sounding rocket.

Public Obsidian Notes Repository

Jul. 2025 – Present

Developed a public-facing repository system that selectively syncs content from a private Obsidian vault using GitHub Actions. Demonstrates experience with automation workflows, version control, and technical documentation practices.

Education

Bachelor of Science (B.S.), Mechanical Engineering

Sept. 2019 – Aug. 2024

I encourage you to check out my website!
dextercarpenter.dev

Skills

Mechanical Design
3D Printing
Composites
Machining
GD&T
Design for Manufacture
Sheet Metal Design

Finite Element Analysis
Cost-Benefit Analysis

PLC Programming
Control System Design
HMI Programming
Digital Twin
Hardware-Agnostic Simulation

Git
Library Development
API Development
CI/CD

Technical Writing
Technical Drawings
Documentation

Microcontrollers
Computer Networking
Virtualization

Control Panel Design

Applications

 Microsoft Office
 Google Workspace
 Monday
 Slack

 Inventor
 Fusion360
 AutoCAD
 Solidworks
 OnShape

 TwinCAT

 GitHub
 Docker
 VSCode
 Arduino

 Bambu

 Obsidian

Programming Languages

 IEC 61131-3 (Structured Text)
 C#/C++
 Python
 MATLAB
 HTML/CSS
 Ruby

References

Nathan Butler, Superior at TAC
Mike Stewart, Flyer of Record & Mentor

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